Cluster Dynamics Codes for Radiation Damage in Nuclear Materials: Xolotl versus RadCluster

Capabilities, Differences, and Potential Applications

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Abstract

Xolotl and RadCluster are two open-source cluster dynamics (CD) codes addressing radiation damage in nuclear materials. Both integrate size-resolved master rate equations for defect clusters, but they differ fundamentally in spatial framework, target material systems, He –vacancy state-space treatment, and computational architecture. This paper provides a side-by-side comparison of the two codes across physics scope, numerical methods, material-specific features, and planned extensions, identifying the application domains where each excels as well as opportunities for combined use.

Contents

1 Introduction

Cluster dynamics (CD) simulation tracks the time evolution of defect cluster populations under irradiation by integrating a coupled set of master rate equations—one per cluster species and size—that describe cascade production, defect diffusion, cluster growth, shrinkage, dissolution, and mutual reactions. Two open-source CD codes occupy non-overlapping niches in the radiation damage modeling landscape:

Xolotl [1] (ORNL, SNL, CEA): A high-performance, spatially resolved advection-reaction-diffusion (ARD) code developed under DOE SciDAC programmes for plasma-surface interaction (PSI) in fusion divertors and fission fuel performance. Current release: v3.3.0 (March 2026). Multiple material networks are provided: PSI/tungsten (He, D, T, V, I), Fe (He, V, I), generic alloys (voids, Frank and perfect loops), Zr (basal/prismatic loops), and nuclear fuels (Xe fission gas).

RadCluster [2] (UCLA, N.M. Ghoniem): A bulk (spatially homogeneous) CD code tailored to *neutron* irradiation of ferritic-martensitic steels, specifically EUROFER97.The code tracks three categories: (1) **active variables**: concentrations with significant time derivatives or strong coupling to the evolving state; (2) **quasi-equilibrated variables**: concentrations that adjust rapidly and remain close to local steady state; and (3) **dormant variables:** concentrations that are negligible or kinetically disconnected over the current time window. The governing equations form a high-dimensional stiff cluster-dynamics system. Although the full state includes ODEs for all admissible defect cluster sizes, only a dynamically changing subset is active at any given time; many species remain near quasi-equilibrium, while others are dormant over long intervals. It is a multiscale cluster-population ODE system with a time-dependent active set. The code implements (i) explicit solute-trapping corrections for the concentrated EUROFER alloy matrix, (ii) mixed 1D/3D transport for glissile SIA clusters, (iii) a hybrid discrete/bin-moment reduction for tractable treatment of large cluster populations, and (iv) two physics-based He–vacancy state-space reductions matched to fission and fusion neutron spectra. Planned extensions add hydrogen as a fourth species and a hybrid deterministic–stochastic solver architecture.

The paper is structured as follows: Section 2 compares the physics scope; Section 3 addresses numerical methods; Section 4 surveys material-specific features; Section 5 provides a comprehensive side-by-side table; Section 6 describes the planned extensions for RadCluster; and Section 7 identifies application domains and opportunities for combined use.

2 Physics Scope

2.1 Spatial Framework

Xolotl solves a one-dimensional advection-reaction-diffusion PDE for each cluster species:

$$\frac{\partial C_i}{\partial t} = \frac{\partial}{\partial x}\left(D_i \frac{\partial C_i}{\partial x}\right) - \frac{\partial}{\partial x}(v_i C_i) + R_i, \tag{1}$$

where D_i is the cluster diffusivity, v_i is the drift velocity (e.g. due to stress or temperature gradients near a surface), and R_i encodes all reaction and source terms. This framework is essential for plasma-facing applications where the He/D/T implantation range (∼1–100 nm) is much smaller than the component thickness and spatial gradients are strong.

RadCluster integrates ordinary differential equations (ODEs) for the bulk cluster concentrations, appropriate for deep-bulk neutron irradiation where spatial gradients are negligible:

$$\frac{dc_n}{dt} = G_n + \mathcal{R}_n[\{c_n\}] - \mathcal{D}_n\, c_n, \tag{2}$$

where G_n is the cascade production rate of size-n SIA clusters, $\mathcal{R}_n$ encodes all cluster–cluster reactions and thermal emissions, and $\mathcal{D}_n$ represents fixed-sink losses to dislocations, grain boundaries, and precipitates.

2.2 Defect Species and Reaction Networks

Table 1 summarises the defect species tracked in each code's reaction networks.

2.3 He–Vacancy State Space

Xolotl tracks every He_xV_y cluster as a distinct species on a 2D composition grid in (x, y) space. The cluster generator produces all energetically allowed compositions up to user-specified maximum sizes; the resulting per-species ODEs are then discretised on the 1D spatial grid. This is the most complete representation of He–vacancy co-evolution but scales as $O(x_{\max} \cdot y_{\max})$ in species count.

RadCluster avoids this combinatorial scaling through two physics-motivated state-space reductions. In both cases the full SIA population and the free-He balance are retained unchanged; only the vacancy-cluster block is compressed. The maximum He loading per cavity scales as

$$\ell_{\max}(m) = \lfloor \alpha_{\mathrm{He}}\, m^{2/3} \rfloor, \quad \alpha_{\mathrm{He}} = 1.5\text{–}2.0, \tag{3}$$

giving a full vacancy–He grid of $n_{VH}^{\mathrm{full}} \sim M^{5/3}$ equations (up to 10^4–10^5 for $M = 10^3$–10^4).

Case 1 — Fusion (mean-field He). Under fusion-relevant He generation (∼10 appm/dpa), He equilibrates on each cavity faster than the cavity grows or shrinks. The 2D (m, ℓ) distribution collapses to two scalars per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading $\bar{\ell}(m)$. System size: $N + 2M + 1$.

Case 2 — Fission (decoupled He inventory). At fission He production rates (∼0.5–1 appm/dpa), most cavities are pure voids and He acts as a weak perturbation. He is tracked as a single scalar total inventory ℓ_{tot}, distributed algebraically across cavity sizes in proportion to their capture cross-sections:

$$\ell(m) = \ell_{\mathrm{tot}}\, \frac{m^{1/3}\, \bar{c}_m}{\sum_{m'} m'^{1/3}\, \bar{c}_{m'}}. \tag{4}$$

Table 1: Defect species tracked in each code. Planned species for RadCluster are shown in green.

Feature	**Xolotl v3.3**	**RadCluster (current + planned)**
Species tracked	He, D, T, V, I (PSI/W network)	SIA clusters, $n = 1, \ldots, I$
	He, V, I (Fe network)	Vacancy/void clusters, $m = 1, \ldots, V$
	V, Void, Faulted, I, Perfect, Frank (Alloy)	He–vacancy clusters, $\ell = 0, \ldots, \ell_{\max}(m)$
	V, Basal, I (Zr network)	Free helium c_h
	Xe, fission-gas pairs (NE/fuel)	Free hydrogen c_H
		H–He–vacancy clusters (m, ℓ, p)
Hydrogen isotopes	D, T explicitly tracked (PSI network)	H planned (planned extension)
He/gas co-production	Implanted from plasma; externally set	Neutron transmutation rate (appm He/dpa); fission/fusion choice
Loop morphology	Generic I (PSI/Fe); Perfect + Frank (Alloy)	$\frac{1}{2}\langle 111 \rangle$ (glissile, 1D glide) and $\langle 100 \rangle$ (sessile), with distinct bias factors
Trap mutation	Yes (W, via He over-pressurisation)	Yes (Fe, He–V clusters; $\Gamma_{\mathrm{TM}} = \nu_0 e^{-E_{\mathrm{TM}}/k_B T}$)
Radiation re-solution	Yes	Yes ($b_0 \, \ell \, \dot{\phi}$ per He atom; $b_0^H \, p \, \dot{\phi}$ for H)

System size: $N + M + 2$.

In both cases, the He gas pressure correction to the vacancy binding energy is retained via the virial equation of state:

$$P_{\mathrm{He}} V = N_{\mathrm{He}}\, k_B T \left[1 + B_2 (N_{\mathrm{He}}/V) + B_3 (N_{\mathrm{He}}/V)^2\right], \tag{5}$$

with $B_2 = 1.67 \times 10^{-29}\,\mathrm{m}^3/\mathrm{atom}$ and $B_3 = 1.84 \times 10^{-58}\,\mathrm{m}^6/\mathrm{atom}^2$ (representative of 600–1000 K).

2.4 Cascade Production Spectrum

Xolotl focuses on ion and plasma implantation scenarios; the displacement source is prescribed from TRIM/SRIM stopping-range calculations. Neutron Frenkel-pair production is supported in some networks but without size-resolved in-cascade cluster distributions.

RadCluster implements an explicit power-law in-cascade cluster spectrum derived from MD simulations of α-Fe and parameterised separately for fission and fusion neutron spectra:

$$\epsilon_m^{(i)} = C_i\, m^{-s_i}, \qquad C_i = \frac{f_i^{\mathrm{cl}}}{\displaystyle\sum_{k=2}^{i_{\mathrm{cascade}}} k^{1-s_i}}. \tag{6}$$

The parameters (Table 2) are grounded in the arc-dpa framework [3] and MD databases for α-Fe.

Table 2: In-cascade cluster production parameters in RadCluster.

Parameter	**Fission**	**Fusion**
Survival efficiency η	0.30	0.28
SIA clustering fraction f_i^{cl}	0.58	0.65
SIA power-law exponent s_i	1.6	1.5
Max SIA cluster size i_{cascade}	20	50
Vacancy clustering fraction f_v^{cl}	0.15	0.20
Vacancy power-law exponent s_v	2.5	2.3
Max vacancy cluster size v_{cascade}	10	20
He co-production (appm He/dpa)	0.5–1	$\sim$10

3 Numerical Methods

3.1 Cluster Size Space Discretisation

Xolotl uses full per-composition resolution up to a maximum cluster size set at initialisation. For PSI applications the $\mathrm{He}_x\mathrm{V}_y$ composition grid may contain $O(10^4)$ species; the

ARD equations for each species are then solved simultaneously on the 1D spatial grid. The PETSc IMEX solvers handle the resulting large, dense system exploiting MPI parallelism and, optionally, GPU acceleration.

RadCluster employs a *hybrid discrete/bin-moment* scheme:

- Cluster sizes $1, \ldots, i_{\text{discrete}}$ are resolved individually (one ODE per size).
- Sizes $i_{\text{discrete}} + 1, \ldots, I$ are grouped into I_{bin} logarithmically spaced bins, each represented by P moments $\mu_k^{(0)}$, $\mu_k^{(1)}$, $(\mu_k^{(2)})$:

$$\mu_k^{(0)} = \sum_{n \in \mathcal{B}_k} c_n, \qquad \mu_k^{(1)} = \sum_{n \in \mathcal{B}_k} n\, c_n. \tag{7}$$

- Three intra-bin shape functions give truncation errors of $O((r-1)^2)$, $O((r-1)^3)$, and $O((r-1)^4)$ for constant, linear, and log-normal closures respectively.

With $I = 1000$, $V = 1000$, $i_{\text{discrete}} = 20$, and bin ratio $r = 1.5$, the system shrinks from ∼2000 discrete ODEs to ∼120–200 moment equations, enabling multi-thousand-second irradiation histories on a laptop.

3.2 ODE/PDE Solver Architecture

Table 3: Solver infrastructure comparison.

Aspect	**Xolotl v3.3**	**RadCluster**
Governing equations	1D ARD PDEs	0D ODEs (bulk)
Time integration	PETSc IMEX (CVODE, TS)	SUNDIALS CVODE BDF, order 1–5
Linear solver	PETSc KSP (GMRES, direct)	Dense / band / GMRES
Parallelism	MPI + CUDA/OpenMP (Kokkos)	OpenMP intra-step (optional)
Conservation diagnostics	Not documented	Frenkel-pair + He balances, δ_{FP}, $\delta_{\text{He}} < 10^{-6}$
Implementation language	C++17	C++17 (solver) + Python (driver)
Build system	CMake + Kokkos	CMake + pip

3.3 Stiffness Management

The stiffness ratio in RadCluster is dominated by the ratio of He to vacancy reaction frequencies:

$$\frac{\omega_h^{\text{eff}}}{\omega_v^{\text{eff}}} \sim 10^4\text{–}10^8, \tag{8}$$

arising from the 0.06 eV vs. 0.67 eV migration barriers of He and vacancies. CVODE with adaptive BDF integrates this efficiently with recommended tolerances $\texttt{rtol} = 10^{-8}$, $\texttt{atol} = 10^{-25}$. Xolotl faces the same stiffness in the reaction block, handled by PETSc's IMEX framework, with the additional challenge of a large spatial grid.

4 Material-Specific Physics

4.1 Xolotl: Tungsten Plasma-Facing Components (PSI Network)

The PSI network is the most mature in Xolotl and is optimised for *tungsten divertor armour* in ITER/DEMO:

- **Multi-species plasma**: simultaneous He, D, and T implantation with coupled $\mathrm{He}_x\mathrm{V}_y$ and H isotope trapping.
- **Trap mutation**: He over-pressurisation drives $\mathrm{He}_x\mathrm{V}_y \to \mathrm{He}_x\mathrm{V}_{y+1} + \mathrm{I}$, creating vacancies and nucleating bubbles.
- **Hydrogen retention**: D and T are trapped at He–vacancy clusters and vacancies; a primary driver for tritium inventory predictions.
- **Depth-resolved profiles**: the 1D ARD framework resolves the implantation zone (1–100 nm) through the Bragg-peak damage profile, capturing surface recombination and outgassing.

4.2 Xolotl: Other Material Networks

The Fe network tracks He/V/I for fission reactor pressure vessel and structural steels with a minimal species set. The Alloy network uses six species (V, Void, Faulted, I, Perfect, Frank loops) for generic fcc/bcc alloys, enabling loop-character evolution. The Zr network models basal and prismatic loops in hexagonal zirconium cladding. The NE network handles Xe/Kr fission-gas retention and grain-boundary release in nuclear fuels.

4.3 RadCluster: EUROFER97 Ferritic-Martensitic Steel

4.3.1 Solute-Modified Diffusivities

EUROFER97 (8.9 Cr–1.1 W–0.47 Mn–0.20 V–0.14 Ta–0.11 C–0.03 N wt%) contains dissolved solutes that act as traps for migrating defects. The effective diffusivity of species α is reduced by trap-limited kinetics:

$$D_\alpha^{\mathrm{eff}} = \frac{D_\alpha^{\mathrm{Fe}}}{1 + \displaystyle\sum_s z_s\, c_s\, \exp(E_b^{\alpha\text{-}s}/k_B T)}. \tag{9}$$

At 300 °C the SIA effective diffusivity is reduced by 2–4 orders of magnitude relative to pure α-Fe, primarily by dissolved C ($E_b^{C\text{-SIA}} \approx 0.45\,\mathrm{eV}$) and N; the effect becomes negligible above ∼600 °C. Vacancy and SIA cluster mobilities are treated with the same framework.

4.3.2 1D Glide of SIA Clusters

Clusters of $n \geq 4$ SIAs adopt $\frac{1}{2}\langle 111 \rangle$ loop configurations and migrate by thermally activated 1D glide with a size-dependent diffusivity:

$$D_n^{\mathrm{1D}} = \frac{3a^2\nu_0}{2\,n^s}\exp\left(-\frac{E_m^{\mathrm{1D}}}{k_B T}\right), \quad E_m^{\mathrm{1D}} \approx 0.03\text{–}0.05\ \mathrm{eV}. \tag{10}$$

In EUROFER the mean free path between Burgers-vector changes is estimated at $L_{\mathrm{eff}} \sim$ 3–30 nm (vs. 100–1000 nm in pure Fe), placing cluster transport firmly in the mixed 1D/3D regime. The mixed-geometry reaction rate with spherical cavities is:

$$\mathcal{K}_{n,m}^{(\mathrm{cav})} = \frac{A_{\mathrm{sph}}\, m^{1/3}\, \omega_n^{\mathrm{1D}}}{1 + B_{\mathrm{rot}}\, \hat{L}^2\, m^{-1/3}}, \quad \hat{L} = L_{\mathrm{eff}}/a, \tag{11}$$

transitioning from 3D kinetics (mono-SIA) to 1D/3D hybrid (clusters $4 \leq n \leq n_{\max}^i$) to sessile ($n > n_{\max}^i$).

4.3.3 Binding Energies

Vacancy binding to voids uses a blended atomistic–continuum model, transitioning from a DFT/MD-calibrated power-law fit at small sizes to the capillary expression $E_b^{\mathrm{cont}}(m) = E_f^v - A_{\mathrm{void}}[m^{2/3} - (m-1)^{2/3}]$ at large m. For He-containing clusters (bubbles), the gas-pressure correction enters through the virial EOS:

$$E_b^v(m,\ell) = E_f^v - \Omega\left[\frac{2\gamma_s}{R(m)} - P_g(m,\ell)\right] + A(m)\, e^{-\lambda(m-1)}, \tag{12}$$

where P_g is computed from the He virial EOS at each time step. Interstitial loop binding uses a blended fit to Osetsky MD data, smoothly transitioning to the Frank-loop continuum expression at $n \approx 25$ SIAs.

4.3.4 Conservation Diagnostics

Two exact conservation identities are enforced at every output step. The *swelling identity* expresses total void swelling as

$$S(t) = S_I(t) + \Delta J^d(t), \tag{13}$$

where $S_I = \sum_n n\, c_n$ is the current SIA loop inventory and ΔJ^d is the cumulative net bias flux absorbed by the dislocation network. A *He conservation* identity similarly tracks the total He inventory. Both normalised diagnostics, δ_{FP} and δ_{He}, should remain below 10^{-6}; values above 10^{-3} signal a coding error.

5 Quantitative Capability Comparison

Table 4 provides a comprehensive side-by-side comparison of current capabilities. Items shown in green denote planned RadCluster extensions.

Table 4: Capability comparison: Xolotl v3.3 vs. RadCluster.

Capability	Xolotl v3.3	RadCluster
Governing equations	1D ARD PDEs ($\partial C/\partial t + \nabla \cdot D\nabla C = R$)	0D ODEs (bulk, spatially uniform)
Target material(s)	W (PSI), Fe, generic alloys, Zr, nuclear fuels	EUROFER97 (9Cr F/M steel)
Irradiation source	Plasma (He/D/T) + ion beams + neutrons	Neutrons (fission and fusion spectra)
Hydrogen isotopes	D, T (PSI network, full ODE)	H planned: adiabatic elimination for $m > m_H^*$
He species treatment	Full He_xV_y composition grid	Mean-field (fusion) or decoupled scalar (fission); virial EOS
Spatial resolution	1D depth profiles (nm–mm)	Bulk (spatially uniform)
Surface/boundary conditions	Yes: implantation profile, recombination, sputtering	Not included
In-cascade cluster spectrum	Generic Frenkel-pair source (no size-resolved cascade input)	Power-law $\epsilon_m^{(i/v)}$; fission/fusion parameterisation (MD-based, arc-dpa)
Solute trapping	Not explicit (material-averaged D)	Yes: Cr, Mn, W, C, N with DFT binding energies; D_α^{eff}
1D SIA cluster glide	Not implemented	Yes: $D_n^{1D} \propto n^{-s}$; mixed 1D/3D cavity reaction
Loop Burgers vector	Alloy: Perfect + Frank; PSI: generic I	$\frac{1}{2}\langle 111\rangle$ (glissile) + $\langle 100\rangle$ (sessile); distinct Z_i
Trap mutation	Yes (W)	Yes (Fe, He-vacancy): $\Gamma_{\text{TM}} = \nu_0 e^{-E_{\text{TM}}/k_B T}$
Radiation re-solution	Yes	Yes: He $b_0 \ell \dot{\phi}$; H $b_0^H p \dot{\phi}$
Cluster size compression	Full per-composition grid (truncated at $N_{\max}$)	Hybrid discrete + log-spaced bin-moments ($P = 1, 2, 3$)
He–vacancy state space	Explicit 2D (x_{He}, x_V) grid	Reduced: Case 1 mean-field or Case 2 decoupled scalar

Continued...

Capability	Xolotl v3.3	RadCluster
Multispecies gas state space	He/D/T grid resolved per composition	(m, ℓ, p) for small clusters; adiabatic H for large clusters
Large-cluster solver	Deterministic per-species ODE on spatial grid	Hybrid: Lie–Trotter det.+stoch. for 4-species planned
Parallel computing	MPI + CUDA/OpenMP (Kokkos)	OpenMP intra-step; single-node (current)
Conservation diagnostics	Not documented	δ_{FP}, $\delta_{\mathrm{He}} < 10^{-6}$; exact swelling identity
Experimental validation	W-PSI benchmarks; FeCrAl benchmarks	EUROFER97 TEM/APT irradiation database
Output observables	Depth profiles of concentration, density, diameter	Loop density/size, void swelling, He content, mean sizes vs. dose
Licence	Open source (GitHub: ORNL-Fusion/Xolotl)	Open source (GitHub: ghoniem/EuroferMicrostructure)

6 Planned Extensions for RadCluster

6.1 Hydrogen as a Fourth Species

The current three-component system (SIA clusters, He–vacancy clusters, free He) will be extended to include hydrogen (H) as an explicit fourth species. This is motivated by fusion-relevant conditions—particularly accelerator-driven systems where the hydrogen transmutation rate can reach ∼1000–1500 appm H/dpa—and by the desire to compute tritium retention in ferritic structural components.

Key physical inputs that distinguish H from He:

- H migration is faster: $E_m^H \approx 0.04$–$0.09\,\mathrm{eV}$ vs. $E_m^{\mathrm{He}} \approx 0.06\,\mathrm{eV}$ and $E_m^v \approx 0.67\,\mathrm{eV}$.
- H binds moderately to vacancies: $E_b^{H\text{-}V} \approx 0.43$–$0.57\,\mathrm{eV}$ for $\mathrm{H}_n\mathrm{V}_1$ with $n \leq 6$.
- H occupies the *surface shell* of He–V bubbles, forming a core-shell He(interior)–H(exterior) structure with $E_b^{H\text{-bub}} \approx 0.4$–$0.8\,\mathrm{eV}$.
- H–SIA binding is negligible ($\lesssim 0.1\,\mathrm{eV}$), so H–SIA clusters are treated as instantaneously dissociating.

The strategy uses two tiers. For small clusters ($m \leq m_H^* \approx 5$), H is tracked explicitly as an additional composition index p, yielding a three-dimensional state (m, ℓ, p) per vacancy-containing species. For large clusters ($m > m_H^*$), H is *adiabatically eliminated*: the quasi-static condition $dc_H/dt = 0$ gives an algebraic H loading $\bar{p}(m, \ell)$ per cluster, reducing the state space back to the (m, ℓ) grid. Six new reaction processes (P9–P14) describe H capture, thermal emission, fixed-sink loss, SIA interaction, H–V recombination, and cascade re-solution.

An exact H conservation law,

$$\frac{dS_H}{dt} = G_H - \mathcal{D}_H \, c_H, \quad S_H \equiv c_H + \sum_{m,\ell,p} p \, c_{m,\ell,p}, \tag{14}$$

analogous to the existing He conservation, provides a numerical accuracy diagnostic for the extended system.

6.2 Hybrid Deterministic–Stochastic Solver

After the H extension, the four-species (I, V, He, H) state space scales as $O(N \cdot M \cdot L_{\mathrm{He}} \cdot P_{\max})$, reaching 10^4–10^5 equations even after the He mean-field and H adiabatic reductions. Two fundamental bottlenecks motivate a hybrid solver:

1. **Large-cluster stiffness**: immobile large voids/bubbles evolve on timescales 10–10^5 s while mobile monomers equilibrate in 10^{-6}–10^{-3} s; implicit ODE solvers require a dense Jacobian of size n_{eq}^2.
2. **Combinatorial explosion**: the explicit (m, ℓ, p) tracking at small sizes is tractable, but the number of large-cluster species is too large for per-species ODEs.

The proposed hybrid couples:

Deterministic block $\mathcal{D}$ (small clusters, $n \leq N^i_{\mathrm{front}}$, $m \leq M_{\mathrm{front}}$): SUNDIALS CVODE with adaptive BDF integrates the full four-species master equations including explicit H tracking, using large clusters only as frozen sink terms. System size $n^{\mathrm{det}}_{\mathrm{eq}} \lesssim 1000$.

Stochastic block $\mathcal{S}$ (large clusters): a particle-based stochastic simulation algorithm (SSA) evolves large loop and void populations as integer particle counts, exploiting the *linearity* of large-cluster reactions in the monomers (fixed by block $\mathcal{D}$). H is treated via the adiabatic loading $\bar{p}(m, \ell)$.

The blocks alternate via a first-order Lie–Trotter operator splitting. The macro-timestep δt and particle count N_{part} are independently tunable, giving direct control over the accuracy–cost trade-off. The expected system-complexity reduction is summarised in Table 5.

Table 5: ODE/particle count comparison for the four-species system. $N = M = 10^3$; $L_{\mathrm{He}} \sim 1.5\, M^{2/3}$; $P_{\max} \sim M^{2/3}$.

Approach	**SIA block**	**Void/He/H block**
Full deterministic (current)	10^3 ODEs	10^3 ODEs (He reduction only)
Full 4-species deterministic	10^3 ODEs	$\sim 10^5$ ODEs
Hybrid det.+stoch. (planned)	$\lesssim 200$ ODEs	$\sim 10^3$–10^4 particles

7 Potential Applications

7.1 Where Xolotl Excels

1. **Tungsten divertor armour (ITER, DEMO)**: Xolotl is the code of choice for predicting He bubble nucleation and growth, surface blistering, and tritium retention in W under mixed He/D/T plasma bombardment. The 1D depth resolution is essential because the He implantation range ($\sim$1–10 nm) is far smaller than the component thickness.

2. **Ion-beam irradiation experiments**: laboratory ion beams produce non-uniform damage profiles matched by the ARD framework with TRIM/SRIM source terms, enabling direct comparison with TEM depth profiles.

3. **Multi-species plasma scenarios**: simultaneous He + D/T bombardment requires a coupled $\mathrm{He}_x\mathrm{D}_y\mathrm{T}_z\mathrm{V}_m\mathrm{I}_n$ state space native to Xolotl's PSI network.

4. **Fission fuels**: the NE network models Xe/Kr fission-gas retention, bubble swelling, and grain-boundary release in UO_2 and metallic fuels.

5. **Zirconium cladding**: the Zr network (basal/prismatic loop evolution) is directly applicable to PWR/BWR cladding growth.

7.2 Where RadCluster Excels

1. **Neutron irradiation of ferritic-martensitic steels for fusion blankets**: EUROFER97 is the primary structural material for the EU DEMO breeding blanket. RadCluster faithfully models neutron cascade damage with size-resolved cluster spectra, He co-production at fusion and fission rates, solute-modified mobilities, and 1D SIA cluster glide—all absent in Xolotl's Fe network.

2. **Long-duration bulk microstructure evolution (0–200 dpa)**: the bin-moment reduction allows integration to full reactor lifetimes with clusters tracked to $N = 1000\,\mathrm{SIA}$, while per-size codes become prohibitively expensive beyond $\sim$100 SIA at long irradiation times.

3. **Fission–fusion spectral comparison**: a single input flag switches the cascade production spectrum, enabling direct comparison of fission test-reactor data with fusion neutron predictions for the same material—essential for the "fission–fusion correlation" problem in DEMO material qualification.

4. **Solute composition studies**: the explicit trapping model enables parametric studies of EUROFER composition variants (W content, Ta vs. Nb additions, dissolved C and N levels) on loop density, void swelling, and He bubble populations.

5. **Accelerator-driven systems (after H extension)**: once the H extension is implemented, RadCluster will model H and He co-evolution in structural steels under the high H transmutation rates ($\sim$1000–1500 appm/dpa) of ADS targets and first-wall materials.

6. **Parameter calibration**: the Python driver with Jupyter notebooks supports systematic parameter scans and Bayesian calibration against the EUROFER97 TEM/APT irradiation database.

7.3 Complementary and Combined Use

The two codes address non-overlapping regimes and can be combined in multi-scale analyses:

- A *first-wall simulation* can use Xolotl near the plasma-facing surface (implantation zone, 1–100 nm) and RadCluster for the bulk structural region ($> 10\,\mu$m) where neutron damage dominates. Large-cluster concentrations from Xolotl can seed initial conditions for the RadCluster bulk run.
- Xolotl's Fe network provides spatially resolved He-bubble distributions near grain boundaries; these supply position-dependent nucleation rates for the homogeneous RadCluster calculation.
- The power-law cascade spectrum of RadCluster—parameterised separately for fission and fusion neutrons—could be incorporated into Xolotl's Fe network to improve its neutron cascade fidelity beyond a simple Frenkel-pair source.
- RadCluster's solute-trapping framework could inform composition- averaged effective diffusivities for use in Xolotl simulations of reduced-activation steels.

8 Summary

Xolotl and RadCluster are complementary codes occupying distinct but adjacent niches in the radiation damage modelling landscape. Xolotl's primary strength is the coupling of 1D spatial resolution, surface physics, and multi-species plasma interaction—indispensable for plasma-facing material design in current and next-generation fusion devices. RadCluster's primary strength is its faithful representation of neutron cascade damage spectra, solute trapping in concentrated alloys, long-time bulk microstructure evolution, and exact conservation diagnostics—essential for structural material qualification for fusion blankets and fission reactors.

The planned extensions to RadCluster—hydrogen as a fourth species with adiabatic elimination for large clusters, and a hybrid deterministic–stochastic solver for the four-species (I, V, He, H) system—will extend its applicability to accelerator-driven systems and DEMO first-wall conditions where both He and H transmutation rates are high. Used together, and connected by shared physical models, the two codes span the full range of neutron and plasma environments anticipated in ITER, DEMO, and ADS structural and divertor components.

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